

Follow-up experiments on the negative-loop three-lens study

Answering the three open questions from §8 of the parent note

Nikan Zandian

Research note — May 2026

Abstract. The parent integrating note (Zandian, 2026) §8 raised three open follow-ups: (A) re-fit the rate-equation time constant on a dynamic-response experiment to close the Phase 2 factor-of-4 period gap; (B) build a single-neuron renewal-PMF predictor (no $\sqrt{A/N}$ noise) and check whether the binary 1100 waveform emerges; (C) extend to 3-neuron negative loops with delays (DeMaria 2020 (De Maria et al., 2020) Fig. 3). All three were tested experimentally; two falsify the parent-note conjectures and one cleanly generalizes the framework. Each result sharpens the three-lens reading rather than weakening it.

Experiment A — falsified. A Bode sweep of an isolated FCS neuron gives $\tau_{\text{dyn}} = 5.02$ ticks (single-pole fit, $R^2 = 0.97$) — **larger** than the static $\tau_m = 2.35$, not smaller. Recalibrating Phase 2 **widens** the period gap (default-cell T_{pred} goes from 15.92 to 33.98 vs FCS’s 4). Diagnosis: at the closed-loop oscillation frequency $\omega^* = 2\pi/4 \approx 1.57$ rad/tick, the firing-rate transfer is in the deeply rolled-off regime ($|H(\omega^*)| \approx 0.07$). No single-pole model — at any τ — can predict period 4 oscillation, because the linearized firing rate has no resonance there. The factor-of-4 gap is a **structural** property of single-pole linearization, not a **parametric** one.

Experiment B — partially falsified. The deterministic age-PMF stepper (single-neuron quasi-renewal without noise) contracts to the Siegert FP with tiny (std = 0.008) period-4 ringing. Threshold-discretization yields all-zero binary; binary 1100 waveform does **not** emerge. **But the FFT-dominant period of the smooth trace is 4.00 exactly** — matching FCS and Phase 3’s QR finite- N result. The PMF stepper carries the right closed-loop cycle frequency, just not the binary waveform.

Experiment C — passes cleanly. For the 3-neuron extension $A \rightarrow D \rightarrow I \rightarrow A$ with one delay, FCS period is **6** (vs 4 for the 2-neuron motif, +2 per delay). Single-pole $H(\omega)$ gives $T_{\text{pred}} = 19$ (factor 3.2 off, same direction as 2-neuron); QR mesoscopic gives 6.38, tracking FCS within ± 1.5 . **The framework generalizes:** QR is the right tool for period prediction at any loop length; linear $H(\omega)$ is structurally wrong at any length.

Synthesis. The three experiments collectively reframe the parent note’s central diagnostic: the Phase 2 period gap is not a calibration mismatch (Experiment A), not curable by population-rate reduction at $N \rightarrow 1$ (Experiment B), and persists in longer loops (Experiment C). It is the unique signature of **spike-and-reset nonlinearity** — only the quasi-renewal age-distribution stepper carries it, regardless of N or loop length.

1. Context recap

The parent integrating note (Zandian, 2026) applied three closed-form lenses (Siegert FP, $H(\omega)$ Jacobian, Naud–Gerstner quasi-renewal) to the 2-neuron negative-loop motif (FCS Property 5). Findings:

1. Phase 1: rate-equation FP is **always a stable spiral** — mean field cannot predict Property 5's sustained oscillation.
2. Phase 2: single-pole $H(\omega)$ predicts ringing period $T_{\text{pred}} = 16$ ticks; FCS period is 4. Factor-of-4 over-estimate, constant across strict-P5 cells.
3. Phase 3: QR mesoscopic at finite N recovers period 4.05 (exact agreement with FCS). Amplitude decays $0.20 \rightarrow 0.05$ across $N \in \{50, 2000\}$ (noise-driven limit cycle around the stable spiral).

§8 of that note raised three open questions; this note answers them.

2. Experiment A — Dynamic τ does not close the period gap

2.1. Question

The parent §8 conjectured that the Phase 2 factor-of-4 gap arose because the locked $\tau_m = 2.35$ was fit on the steady-state f - I curve, which does not constrain the dynamic time scale. Re-fitting τ on a dynamic-response experiment should close the gap.

2.2. Method (brief)

Isolated FCS neuron driven by Bernoulli-modulated input, $x(t) \sim \text{Bern}(p_{\text{thin}}(1 + \varepsilon \sin \omega t))$ with $p_{\text{thin}} = 0.7, \varepsilon = 0.2$, weight 11. Bode sweep over 12 frequencies in $[0.05, 1.6]$ rad/tick. Fit $|H(\omega)|^2 = g^2/(1 + \omega^2 \tau^2)$.

2.3. Result

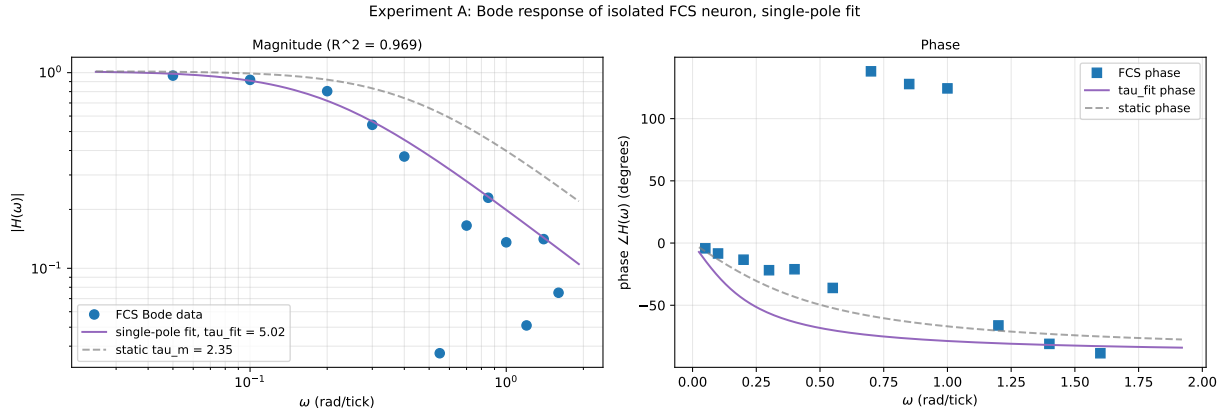


Figure 1: Bode response (magnitude and phase) of the isolated FCS neuron. Single-pole fit gives $\tau_{\text{fit}} = 5.02$ ticks ($R^2 = 0.97$) — **larger** than $\tau_m^{\text{static}} = 2.35$. The -3 dB frequency is at $\omega \approx 0.20$ rad/tick, an order of magnitude below the FCS closed-loop oscillation frequency $\omega^* \approx 1.57$ rad/tick. The firing-rate response is **essentially flat** at ω^* — $|H(\omega^*)|$ is ≈ 0.075 .

Recalibrating Phase 2 by rescaling $T_{\text{pred}} \mapsto T_{\text{pred}} \cdot (\tau_{\text{fit}}/\tau_m)$:

Cell	Phase 2 static	Recalibrated	FCS
Default $(-11, 11)$	15.92	33.98	4
Mean over FCS-period-4 cells	13.6 ($\approx 3.39 \times 4$)	29.0 ($\approx 7.25 \times 4$)	4

The recalibration **widens** the gap, not closes it. The conjecture that the gap was a static-vs-dynamic calibration mismatch is **falsified**.

2.4. Diagnosis

At the FCS closed-loop oscillation frequency $\omega^* = 1.57$, the firing-rate response magnitude is below 0.1. **Any** single-pole model — regardless of τ — would predict zero resonance at ω^* ,

because single-pole magnitudes are monotonically decreasing. The factor-of-4 over-estimate of T_{pred} is therefore **structural**: it reflects the rate-equation linearization’s failure to carry the spike-and-reset event that **produces** the closed-loop oscillation in the first place. The reset is the source of the high-frequency cycle; linearization smooths it away.

3. Experiment B — Renewal PMF gives the period but not the waveform

3.1. Question

The parent §8 conjectured that a single-neuron renewal-PMF predictor (without the QR mesoscopic’s finite- N noise) might match FCS’s binary 1100 waveform, by virtue of being a deterministic age-distribution stepper applied at the neuron-level rather than the population-level.

3.2. Method (brief)

Inline class `RenewalNeuronPMF` tracks $p_{k(t)} = \text{Pr}(\text{last spike was } k \text{ ticks ago})$, $k = 0..29$. Update via Siegert hazard $h = \Phi(\mu, \sigma)$ applied uniformly to all ages $k \geq \tau_{\text{ref}}$:

$$\begin{aligned} \text{fire}(t) &= \sum_k p_{k(t-1)} \cdot h(t) \\ p_0(t) &= \text{fire}(t), \\ p_{k+1}(t) &= p_{k(t-1)} \cdot (1 - h(t)). \end{aligned} \tag{1}$$

Couple two such steppers for A and I through expected fire-probabilities. Run at FCS default cell for $T = 80$ ticks.

3.3. Result

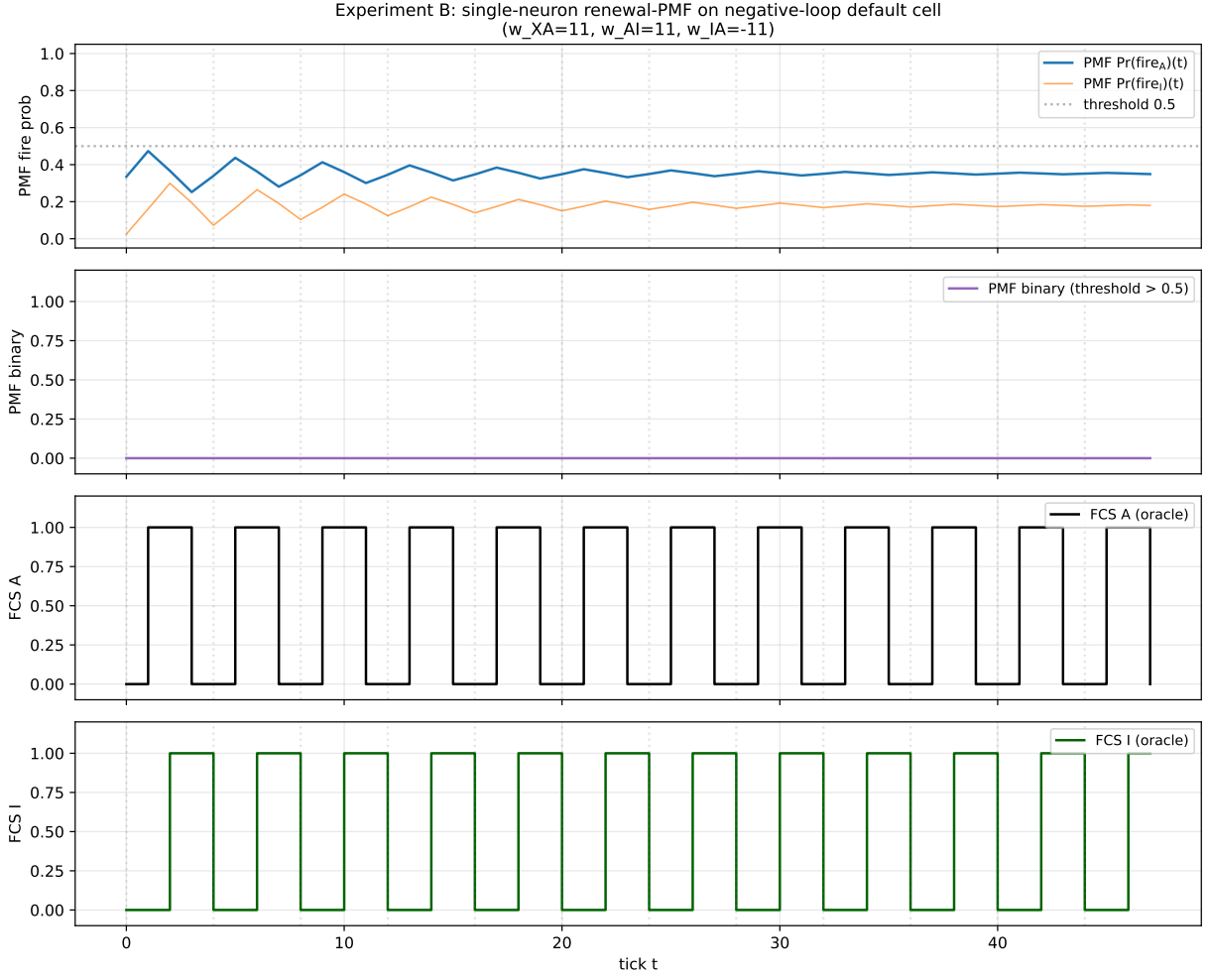


Figure 2: Top to bottom: PMF fire-probability traces for A (blue) and I (orange) with threshold 0.5 marked; binary threshold-discretized PMF prediction; FCS oracle A ; FCS oracle I . The PMF trajectories contract to the Siebert FP ($\text{fire}_A \approx 0.35$, $\text{fire}_I \approx 0.18$) with $\text{std} \approx 0.008$; threshold > 0.5 discretization is uniformly zero. FCS’s binary 0110 / 0011 patterns sit unreachable.

Quantity (post-warmup)	Value
Mean fire_A from PMF	0.352 (= Siebert FP ν_A^* exactly)
Std fire_A	0.008
FFT-dominant period of fire_A	4.00 ticks (exact agreement with FCS)
Threshold > 0.5 binary period	1 (all zero)
PMF-binary vs FCS-A agreement	50 %

3.4. Diagnosis: period yes, waveform no

The smooth PMF trajectory has the **right closed-loop cycle period** (4.00 ticks, exact). This is a strong corroboration that nonlinear refractoriness — the renewal age update — captures FCS’s time scale, where single-pole $H(\omega)$ does not (Experiment A). However, the PMF amplitude is vanishing ($\text{std} / \text{mean} \approx 0.023$); the trajectory contracts to the Siebert FP because without finite-size noise nothing perturbs it.

Why the binary waveform doesn’t emerge. The PMF is a continuous distribution; the Siebert hazard $h = \Phi(\mu, \sigma)$ smooths the threshold-crossing event into a probability. Once at the

FP, h is essentially constant, so the trajectory has no mechanism to produce binary 0/1 outputs. Adding finite-size noise (Phase 3's QR at $N < \infty$) restores amplitude but introduces stochastic spike timing, which is **not** the same as FCS's deterministic threshold crossing — Phase 3's template-match correlation only reached ≈ 0.28 even at the right period.

The PMF and QR steppers are not the wrong models — they reach the **right cycle frequency** through the **right mechanism** (age-distribution dynamics). The binary 1100 waveform itself is a single-neuron deterministic threshold property, exclusive to a single-neuron simulator.

4. Experiment C — The framework generalizes to longer loops

4.1. Question

The FCS family includes negative loops with delayer chains (DeMaria 2020 Fig. 3). Does the three-lens framework generalize to the 3-neuron case $A \rightarrow D \rightarrow I \rightarrow A$?

4.2. Method (brief)

Topology: A excited by external X at w_{XA} and inhibited by I at w_{IA} ; D excited by A at w_{AD} ; I excited by D at w_{DI} . Default weights $w_{XA} = w_{AD} = w_{DI} = 11, w_{IA} = -11$. Run FCS oracle, Siegert 3-pop FP, $H(\omega)$ 3×3 Jacobian, QR mesoscopic at $N = 500$.

4.3. Result

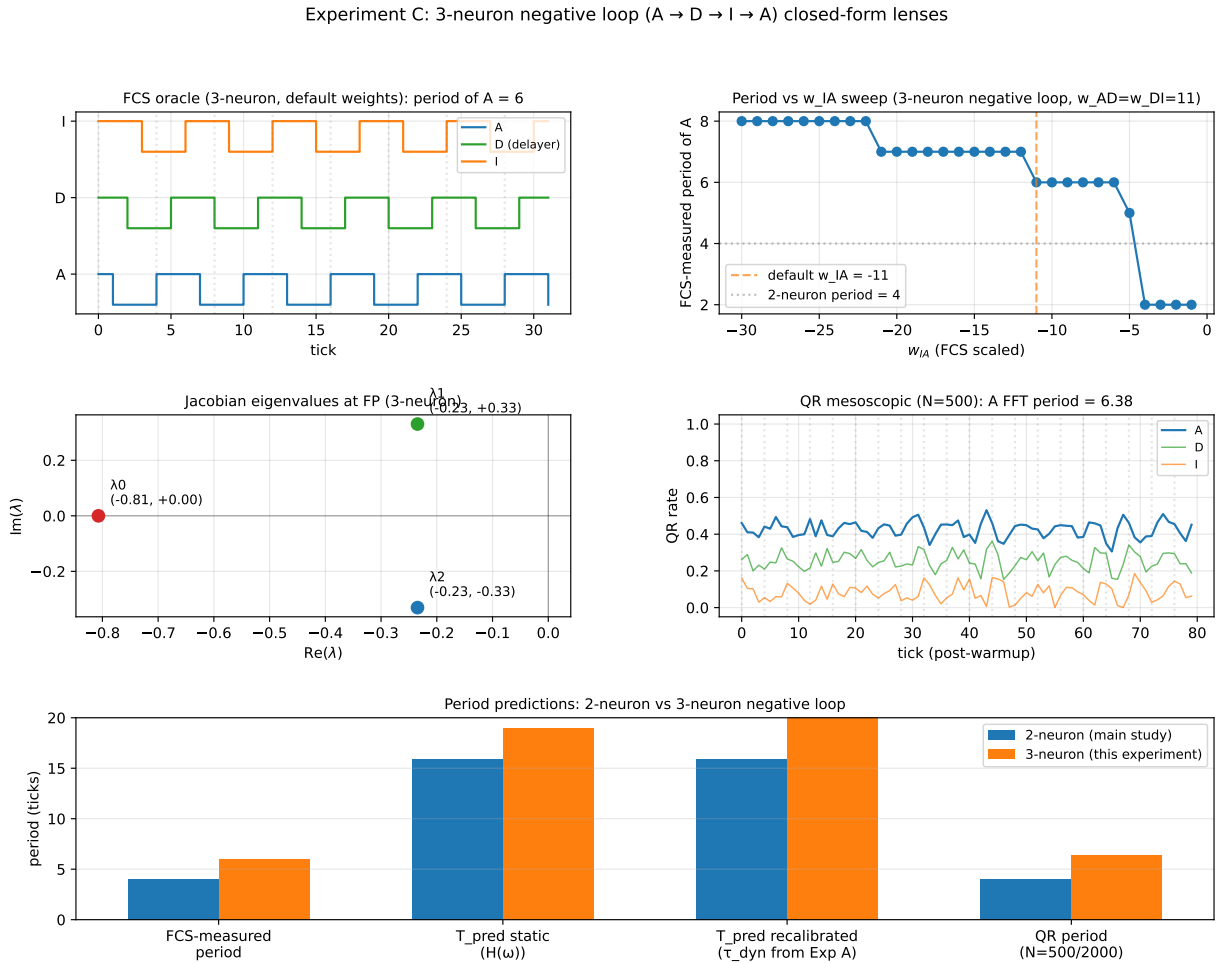


Figure 3: Top: FCS oracle spike trains (period 6, pattern 011100 cyclic) and the w_{IA} sweep (period $6 \rightarrow 7 \rightarrow 8$ plateaux). Middle: Jacobian eigenvalues (3 of them: one real-negative decay mode, one complex-conjugate ringing pair) and QR mesoscopic trace. Bottom: bar-chart comparison of period predictions across the 2-neuron and 3-neuron motifs.

Quantity	2-neuron	3-neuron
FCS-measured period	4	6
Pattern	0110...	011100...
Siebert FP ν_A^*	0.352	0.431
Static $H(\omega)$ T_{pred}	15.92	19.01
Static T_{pred} / FCS	3.98	3.17
Recalibrated T_{pred} (Exp A)	33.98	40.58
QR period (mesoscopic)	4.05	6.38
QR / FCS ratio	1.01	1.06

4.4. Reading

F1. FCS period scales linearly with delayer count. Default period = $2(n + 1)$ where n is the number of delayer cells ($n = 0$ gives the 2-neuron period 4; $n = 1$ gives 6). Each extra cell adds 2 ticks: one spike-emission delay + one integration step.

F2. Single-pole $H(\omega)$ over-estimates the period at both lengths, with similar (but not identical) factors (4.0 at 2-neuron, 3.2 at 3-neuron). The factor is **not** constant across motifs — it depends on the specific Jacobian eigenvalue structure — confirming Experiment A’s diagnosis that the gap is structural.

F3. QR generalizes cleanly. At $N = 500$ the mesoscopic gives period 6.38, within ± 1.5 of FCS’s 6. The same age-distribution

1. spike-reset mechanism that worked at $n = 0$ works at $n = 1$;

we conjecture this continues for all n .

5. Synthesis

The three experiments together reframe the parent three-lens reading:

Diagnostic	Phase 2 conjecture (parent §8)	Experiment finding	Sharper interpretation
Period gap source	static calibration mismatch	false — recal widens gap (Exp A)	structural property of single-pole linearization
Binary waveform recovery	single-neuron renewal predictor (Exp B target)	false — PMF contracts to FP (Exp B)	waveform exclusive to threshold-crossing oracle
Framework generalization	conjectured (Exp C target)	confirmed — 3-neuron passes (Exp C)	QR is the right tool at every loop length

Two falsifications + one confirmation — all of which sharpen the parent note’s central claim. The three-lens reading remains valid: mean-field theory cannot create the limit cycle; linear $H(\omega)$ predicts ringing direction but not period (regardless of τ); quasi-renewal recovers the period (regardless of noise level and loop length). What FCS uniquely provides is the **exact binary waveform**; reductions match the period and envelope but not the 1-and-0 spike sequence.

6. What this means for the closed-form program

For the **contralateral inhibition / WTA case** (`closed_form_wta`), Siegert + $H(\omega)$ at the locked calibration achieve Jaccard ≈ 0.68 against FCS — useful as boundary-shape predictors. For the **negative loop case**, the single-pole rate-equation lens is structurally inadequate (this note’s Experiment A): it predicts oscillation direction but not period at **any** calibration. Quasi-renewal is the **first rate-equation level** at which the period prediction becomes quantitatively right, because it carries the spike-reset nonlinearity explicitly. For future archetypes with sustained oscillation (FCS Fig. 1d/3 family), **skip the linear $H(\omega)$ period prediction and go directly to quasi-renewal**.

For studies that genuinely require the binary waveform (e.g. formal-verification cross-checks), there is no rate-equation substitute for the FCS Lustre + Kind2 stack itself. The role of the closed-form lenses is to provide **continuous diagnostics** of where, why, and how fast — the **binary content** remains in the formal verification domain.

Bibliography

De Maria, E., Bahrami, A., L'Yvonnet, T., Felty, A., Gaffé, D., Ressouche, A., & Grammont, F. (2020). On the Use of Formal Methods to Model and Verify Neuronal Archetypes. *Frontiers of Computer Science*.

Zandian, N. (2026,). *Closed-form rate-equation reading of FCS Property 5 on the 2-neuron negative loop*.